Project Ideation and Problem Statement

Detailed project document · Prepared for project completion and review

Project reference information

Item	Known project detail
Project	Health360 Clinical & Healthcare Risk Analysis
Dataset size	5,000 patient/person records
Dataset fields	17 columns; exact field names and definitions must be copied from the source dataset's header row/data dictionary
Analytics tool	Tableau
Visual assets	9 worksheets/visualizations and a dashboard; a Tableau story for Patient Risk & Lifestyle Insights was also part of the project workflow
Repository	https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git
Website	https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/
Scope caution	This is an educational data-visualization project. It is not a clinically validated diagnostic, prognostic, or treatment system.

Executive summary

Health360 Clinical & Healthcare Risk Analysis is an educational analytics project that organizes a healthcare-related dataset into interactive Tableau views. Its purpose is to help a viewer explore patterns in patient/person risk indicators and lifestyle-related variables, understand how measures vary across the available records, and communicate observations in a structured visual format. The project contains a dataset described as 5,000 records and 17 columns. The exact meaning of each field must be taken from the original dataset documentation rather than inferred from its name.

Problem statement

Healthcare datasets can be difficult to interpret when information is presented only as rows in a spreadsheet. A user may need to compare groups, inspect distributions, notice data-quality limitations, and trace how an observation was produced. Health360 addresses this presentation problem by organizing the data into a dashboard and story. The project should not claim that visualization alone improves clinical outcomes or that associations found in the data establish cause and effect.

Proposed solution

Prepare and validate the dataset; define each measure and dimension; construct a set of Tableau worksheets; assemble them into a dashboard; use a story to sequence selected insights; and publish the views through Tableau Public or the approved hosting environment. A project website provides context and navigation to the published visualizations. Each conclusion should be linked to a specific chart, filter state, and data subset.

Intended users

Potential audiences include students, instructors, data-analytics reviewers, and readers interested in learning how healthcare-related data can be explored visually. Any use by clinicians or patients must be framed as exploratory only unless clinical validation, governance, and appropriate approvals are independently established.

Scope and exclusions

Included: data inventory, data cleaning, exploratory analysis, visualization design, dashboard/story development, publication, testing, and documentation. Excluded unless separately supported: clinical diagnosis, personalized treatment advice, prediction of an individual patient's outcome, causal claims, or claims that the dataset represents all patients or a general population.

Problem statement — final form

How can a healthcare-related dataset of 5,000 records and 17 fields be cleaned, described, and presented through an interactive Tableau dashboard and story so that users can explore available risk and lifestyle indicators while understanding the limits of the data?

Deliverables

A documented dataset and data dictionary; reproducible cleaning log; Tableau workbook containing worksheets and dashboard; a narrative story; a website linking to the visualizations; a test record; a final report and demonstration materials.

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Objectives and Success Criteria

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Primary objective

Build an understandable and testable exploratory analytics product for the Health360 dataset, with traceable metrics and clear limits on interpretation.

Specific objectives

1. Inventory all 17 fields and document their types, units, and meaning. 2. Check the 5,000-record dataset for missing values, duplicates, inconsistent categories, and invalid ranges. 3. Define every calculated field and aggregation used in Tableau. 4. Create nine purposeful worksheets/visualizations that answer distinct analytical questions. 5. Assemble the relevant views into a dashboard with clear titles and usable interactions. 6. Build a story that explains selected observations in a logical order. 7. Publish and test the views and website links. 8. Document limitations, data provenance, and the validation process.

Success criteria

Data integrity: the source row count and post-cleaning row count are recorded; changes are explainable; field definitions are available. Calculation correctness: key counts, averages, rates, and percentages are checked against independently calculated examples. Visualization quality: every chart has a question, title, labels, units where applicable, and a reason for the chosen encoding. Usability: filters and actions work as expected; the dashboard is legible on a typical laptop and mobile screen. Reproducibility: another reviewer can identify the source file, workbook, published links, and transformations.

Acceptance criteria

The project is ready for review when all workbook worksheets render, dashboard controls have been tested, story points follow the intended sequence, links are accessible to the intended audience, all claims in the report can be traced to the dataset or a citation, and known limitations are stated. A pass/fail record should be completed rather than assuming success.

Out-of-scope criteria

Do not present a risk score as clinically validated unless the methodology, outcome definition, validation cohort, calibration, and approval are documented. Do not infer patient identities or expose personally identifiable information. Do not state that a factor causes a health outcome based only on a visual association.

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Stakeholders and User Personas

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Stakeholder register

Project owner/student: maintains scope, source files, workbook, website, and documentation. Faculty/project evaluator: checks the problem definition, method, implementation, and evidence. Data steward/source publisher: provides data definitions and usage terms. Viewer/reviewer: explores charts and checks whether conclusions are understandable. If any real healthcare organization is involved, its privacy, security, and clinical governance stakeholders must be identified before use.

Primary persona

Persona: student data analyst. Goal: explain patterns in a healthcare-related dataset through a concise visual narrative. Needs: clear metric definitions, visible data scope, meaningful filters, consistent color use, and a report that distinguishes observation from interpretation. Pain points: ambiguous field names, unverified assumptions, overcomplicated charts, and dashboard elements that load slowly or do not fit mobile screens.

Secondary persona

Persona: instructor or project evaluator. Goal: assess whether the project is reproducible and analytically responsible. Needs: source provenance, a data dictionary, transformation notes, workbook structure, validation evidence, accessible links, and a transparent limitations section.

User tasks

A user should be able to identify the project purpose, determine which data is included, read the definitions of key measures, compare groups shown in the charts, apply supported filters, reset the view, move through the story, and find the source/limitations. The interface should not imply that a viewer can use the charts to diagnose a person.

Privacy and accessibility

Use only data the project is authorized to process. If records contain identifiers, remove or protect them before publication and verify that combinations of fields cannot reveal identities. Use descriptive titles, sufficient contrast, non-color cues, legible labels, keyboard-friendly website navigation, and meaningful alt text for static images.

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Functional Requirements

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FR-01: data inventory

The project shall record source file name, publisher, retrieval date, license/terms, record count, field count, and the exact 17 column names. Acceptance test: compare the documented inventory to the actual source file and record any mismatch.

FR-02: data validation

The project shall check data types, missingness, duplicates, inconsistent category spellings, invalid values, and unit consistency. Each correction or exclusion must be recorded. Acceptance test: produce a before/after summary and verify that no change is unexplained.

FR-03: metric definitions

Each measure used in Tableau shall have a documented definition, aggregation, denominator where applicable, unit, and handling of null values. Percentages must specify the population included in the denominator. Acceptance test: manually recalculate sample cases.

FR-04: worksheets

The workbook shall contain nine purposeful worksheets/visualizations as stated in the project workflow. For each worksheet, document the analytical question, dimensions, measures, filters, encoding, and expected interpretation. The exact worksheet titles and measures must match the actual workbook; do not invent titles in the final report.

FR-05: dashboard

The dashboard shall combine selected worksheets, provide a clear reading order, use consistent formatting, and expose only meaningful filters/actions. Acceptance test: test each filter with multiple selections, verify cross-filtering if enabled, and test reset behavior.

FR-06: story

The story shall sequence views or annotations so that each point adds context. Every claim must be supported by the displayed data. Acceptance test: review story point order and confirm that each point can be understood without unsupported assumptions.

FR-07: publication and website

The published dashboard/story links and website shall open for the intended audience. If public sharing is used, confirm the data is approved for public release. Acceptance test: open links in a signed-out/private browser session and test on desktop and mobile.

FR-08: documentation

The project shall include a final report, data dictionary, cleaning log, test record, references, demonstration script, and a handoff inventory. Acceptance test: check that the filenames, workbook version, and URLs agree across documents.

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Nonfunctional Requirements

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Usability

The site and dashboard should use a predictable navigation structure, concise headings, descriptive chart titles, readable text, and consistent formatting. Users should be able to distinguish filters from informational labels and understand when a view has been filtered.

Performance

Define a practical target after testing in the intended environment. Record cold-load and repeat-load observations for the dashboard, story, and website. Tableau Public performance depends on network conditions, workbook complexity, browser behavior, and the service; do not promise a fixed load time without measurement. Avoid loading duplicate Tableau scripts and test embedded views independently.

Privacy and security

Use minimum necessary data, protect identifiers, follow the source license and institutional rules, and do not publish sensitive records without authorization. If the source is synthetic or de-identified, describe that only if the source documentation confirms it. Do not assume anonymization merely because names are absent.

Reliability

Document known data-quality issues, workbook version, publication date, and links. Test the website after publication. Keep a local copy of the workbook and source data where permitted. Record failures and the conditions under which they occurred.

Compatibility and accessibility

Test at least one desktop browser and a mobile browser. Verify that embedded views fit their container and that the user can scroll through long visualizations. Use accessible page headings, contrast, keyboard navigation, and text explanations for visual content.

Maintainability

Use clear filenames, documented calculations, stable field names, and a change log. When changing data or workbook calculations, rerun relevant tests and update the report and data dictionary.

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Dataset Inventory and Data Dictionary

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Known inventory

The Health360 project has been described as using 5,000 person/patient records and 17 columns. This is the known project-level inventory. The exact dataset title, publisher, retrieval date, license, column names, definitions, units, and outcome labels must be copied from the source file and its documentation. This template intentionally does not fabricate those details.

Data dictionary procedure

Create one row per source column with: exact field name; plain-language definition; storage type; semantic type; unit; valid range or categories; missing-value representation; privacy classification; source or derivation; transformation rule; Tableau usage; and quality-check result. Preserve the original spelling in a separate field if a cleaned name is used in Tableau.

Data-quality checks

Record total records before and after cleaning; exact duplicate count; null count and percentage by field; category cardinality; invalid ranges; date/time coverage if relevant; and any inconsistent units. Decide how to handle each issue before using it in a metric. Avoid silently replacing missing values with zero because zero may be a valid value.

Derived fields

For each derived field, record the formula, input columns, assumptions, unit, null handling, and a manually verified example. If a risk category is provided by the source, document the source definition. If it is created by the project, label it as a project-defined grouping and do not present it as a clinically established threshold.

Privacy and provenance

Record whether the data is public, synthetic, de-identified, or restricted only when supported by source documentation. Record the license and permitted uses. Remove direct identifiers from any public-facing extracts and check re-identification risks before publishing.

Field inventory worksheet

Fill this table from the actual 17 columns: Field name | Definition | Type | Unit/categories | Missing values | Transformation | Dashboard use | Validation evidence. Do not finalize the report until all 17 fields have been reviewed.

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Dashboard Design Specification

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Design goal

The dashboard should provide an at-a-glance overview of the Health360 dataset and let users explore the available indicators. The final design must reflect actual workbook content; the following specification describes design checks, not a claim that every element is already implemented.

Layout hierarchy

Place a concise title and scope statement first. Group related metrics and charts. Use a consistent grid, aligned edges, adequate spacing, and a clear reading order. Put global filters in a predictable location and explain whether they affect all charts or only selected views. Avoid excessive KPI cards that repeat the same information.

Chart specification

For each of the nine worksheets, document: worksheet title; question; chart type; dimensions; measures; aggregation; filters; sorting; color/size/position encoding; tooltip fields; expected user action; and known limitation. Select chart types based on the analytical task rather than decoration. Use bars for comparisons, lines for ordered time series, distributions for spread, and scatter plots for relationships when appropriate.

Metric and filter rules

A metric card must state its name, unit, aggregation, and population. Percentages must state the denominator. Filters should include sensible defaults, clear labels, and a reset option where possible. Avoid filters that leave the user with no visible explanation when they produce an empty result.

Accessibility and responsive design

Use adequate contrast and avoid relying on red/green alone. Keep chart labels readable. On small screens, prefer vertical stacking or a clear scroll area over shrinking charts until labels become unreadable. Test the actual published dashboard on mobile and desktop.

Dashboard QA checklist

Confirm all worksheets load; titles match the report; units and formats are consistent; filters produce expected results; tooltips are meaningful; sorting is correct; nulls are handled intentionally; dashboard actions are documented; the layout does not hide key information; and the published URL works for a signed-out viewer.

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Story Design and Narrative

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Narrative purpose

The Health360 story, referred to in the project workflow as Patient Risk & Lifestyle Insights, should guide the viewer through a small number of evidence-supported observations. It should not simply duplicate the dashboard. The narrative should establish scope, present a pattern, explain what can be inferred, and state the limitations.

Recommended sequence

Point 1 — Context: explain dataset coverage, record count, and the meaning of key fields. Point 2 — Overview: present a broad distribution or summary that is actually available in the workbook. Point 3 — Comparison: compare groups using a documented metric and denominator. Point 4 — Relationship: show an association only if supported by the selected variables and sample. Point 5 — Interpretation: explain plausible implications without asserting causation. Point 6 — Limitations: disclose data coverage, missingness, possible confounding, and lack of clinical validation.

Evidence rules

Each story point should include a clear title, a chart that supports the title, a brief annotation, and the filter state or population covered. A title should not claim more than the chart demonstrates. Avoid causal terms such as causes, leads to, or prevents unless the research design supports causality.

Writing pattern

Use this pattern for each insight: Observation — what the chart visibly shows; Context — what group/time/metric is shown; Interpretation — what the pattern may suggest; Caveat — what cannot be concluded from the available data. If the evidence is weak or unclear, state that instead of forcing a dramatic conclusion.

Story review checklist

Check the order of story points, spelling, metric consistency, filter state, axis units, and link availability. Ensure the story and dashboard do not use contradictory definitions. Verify that screenshots or exported images correspond to the current published workbook version.

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Project Plan and Milestones

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Work breakdown

Workstream A: source acquisition and provenance. Workstream B: field inventory and cleaning. Workstream C: exploratory analysis and calculation validation. Workstream D: Tableau worksheet creation. Workstream E: dashboard assembly. Workstream F: story writing. Workstream G: website integration and publication. Workstream H: testing and documentation. Workstream I: demonstration and final handoff.

Milestone gates

M1 Scope approved: problem, objectives, and boundaries documented. M2 Data ready: field dictionary and quality checks completed. M3 Analysis ready: worksheets and calculations validated. M4 Visualization ready: dashboard and story reviewed. M5 Publication ready: links, permissions, and embeds tested. M6 Submission ready: report, source references, screenshots, and demo materials checked.

Schedule table

For every task, record planned start, planned finish, actual finish, owner, dependency, effort, status, and evidence link. Dates should be entered by the project owner; this document does not invent deadlines or claim tasks are complete.

Risk register

Risk: data definitions unclear — mitigation: inspect source documentation and avoid guessing. Risk: field names change — mitigation: record source version and retest calculations. Risk: Tableau view is private — mitigation: confirm intended sharing permissions. Risk: embed spins — mitigation: test direct view, network, browser, and embed separately. Risk: unsupported interpretation — mitigation: use evidence review and explicit limitations.

Change control

When scope, source data, calculations, or published URLs change, record the change, reason, date, impact, approval, and tests to rerun. Keep an immutable source copy when permitted and label published versions clearly.

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Test Plan and Quality Checklist

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Testing approach

Testing should cover data quality, calculation correctness, worksheet behavior, dashboard interactions, story consistency, website rendering, and publication access. Record actual outcomes; a checklist is not evidence of a passed test until it is executed.

Data tests

T01: source row count is recorded as 5,000 only if verified against the actual file. T02: 17 source columns are enumerated. T03: field types and units are reviewed. T04: duplicate records are measured and handled by documented rules. T05: missingness is quantified. T06: invalid ranges and inconsistent categories are reviewed. T07: cleaning changes are reproducible.

Calculation tests

T08: independently calculate selected counts and averages on a small sample. T09: verify denominator and null handling for percentages. T10: compare Tableau totals with a source-file pivot or independently computed result. T11: test filters against known subsets. T12: ensure derived categories match documented boundaries.

Visualization tests

T13: every chart has a meaningful title and labels. T14: axes and units are correct. T15: sorting and aggregation are expected. T16: tooltips show appropriate context. T17: filters/actions work and can be reset. T18: nulls and empty results are handled. T19: story sequence and annotations match the actual charts.

Publication tests

T20: open the dashboard directly while signed out. T21: open the story directly while signed out. T22: open the website in desktop Chrome. T23: test the website on mobile. T24: confirm no view is obscured or cropped. T25: test after a clean redeployment and record the tested URLs and date.

Test evidence table

For each test record: ID, precondition, steps, expected result, actual result, pass/fail, defect reference, tester, and date. If a test is not applicable, explain why rather than marking it passed.

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Final Project Report — Detailed Draft Structure

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Abstract

Health360 Clinical & Healthcare Risk Analysis is an exploratory data-visualization project built around a healthcare-related dataset described as containing 5,000 records and 17 fields. The project workflow uses Tableau worksheets, a dashboard, and a story to present the data for interactive exploration. The report should state the verified source, preparation method, actual findings, and limitations. Replace this summary with a final abstract after the results have been verified.

Introduction and problem

Explain why tabular data can be difficult to compare and how a dashboard can support structured exploration. Define the target audience and clarify that the project is educational and exploratory, not a clinically validated decision-support system.

Dataset and preparation

Document source name, publisher, access date, license, exact field list, row count, missingness, duplicate handling, transformations, and privacy considerations. Include a data dictionary and a cleaning log. Do not publish sensitive records without authorization.

Methodology

Describe each analytical question, metric definition, aggregation, grouping, and visualization choice. Explain how calculations were validated. Distinguish descriptive analysis from inferential analysis; do not claim causality based on correlations or group comparisons alone.

Implementation

Describe the Tableau workbook, nine worksheets/visualizations, dashboard, story, and project website. Include actual screenshots and published URLs in the final report. The repository is https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git and the website is https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/; verify both before submission.

Results and discussion

Insert only findings directly supported by the actual charts and source data. For each finding, identify the relevant population, metric, denominator, chart, and filter state. Discuss alternative explanations and data limitations. This template does not invent result values because the exact column definitions and chart outputs are not included here.

Testing and evaluation

Summarize executed data checks, calculation reconciliations, interaction tests, accessibility review, browser tests, and publication checks. Include failures and fixes rather than reporting only successful tests.

Conclusion and future work

Summarize what the project delivered, the questions it helps explore, and its limitations. Future work may include improved documentation, more robust data validation, accessibility refinements, a reproducible refresh pipeline, or validation with an appropriately governed dataset.

References and appendices

Cite the dataset, documentation, methods, Tableau documentation, and any external research used. Append the field dictionary, test results, calculation definitions, screenshots, and a versioned asset inventory.

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Data Preparation and Cleaning Log

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Scope caution	This is an educational data-visualization project. It is not a clinically validated diagnostic, prognostic, or treatment system.

Purpose and reproducibility

This log records how the original Health360 dataset was transformed into the data used in Tableau. Preserve the original source file where permitted and never overwrite it with a cleaned copy. Record each operation so a reviewer can reproduce it.

Initial inspection

Record source filename, file type, size, date received, source URL, license, number of rows, number of columns, encoding, delimiter if CSV, and any data dictionary. The project-level description is 5,000 records and 17 columns; confirm those counts against the actual source file before using them as final results.

Cleaning workflow

Step 1: inspect field names and types. Step 2: quantify missingness. Step 3: check duplicate rows and any available record identifiers. Step 4: normalize category spelling only where values clearly represent the same category. Step 5: validate units and plausible ranges against the source documentation. Step 6: decide how nulls are handled for each calculation. Step 7: document exclusions and compare pre/post counts. Step 8: export or connect the validated data to Tableau.

Transformation log format

For every transformation record: ID; date; source column(s); original issue; rule applied; number of rows affected; reason; before/after example; validation test; and author. If no transformation was applied to a field, record that it was inspected and retained unchanged.

Derived fields and Tableau calculations

Record exact formula, aggregation, units, rounding, null handling, and sample test for each derived field. Use the same definition in the workbook, dashboard labels, and report. Never use a project-defined risk grouping without explaining that it is a project rule rather than an established clinical threshold.

Reproducibility checklist

A second analyst should be able to identify the original file, reproduce the cleaning rules, rebuild key metrics, and reconcile the final record count. Store versions and test evidence in the project repository where permitted.

Completion and review record

Prepared by: _____ Date: _____

Reviewed by: _____ Status: Draft / Reviewed / Approved

Document-control note: Update this file when the dataset, workbook, published views, or website changes. Record version/date and keep claims tied to evidence. This detailed document uses known project context and explicitly flags details that must be verified from the actual source files.

Findings, Interpretation, and Limitations

Detailed project document · Prepared for project completion and review

Project reference information

Item	Known project detail
Project	Health360 Clinical & Healthcare Risk Analysis
Dataset size	5,000 patient/person records
Dataset fields	17 columns; exact field names and definitions must be copied from the source dataset's header row/data dictionary
Analytics tool	Tableau
Visual assets	9 worksheets/visualizations and a dashboard; a Tableau story for Patient Risk & Lifestyle Insights was also part of the project workflow
Repository	https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git
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Evidence-first reporting

A finding must be an observation from a specific chart or calculation. State the measure, population, denominator, filters, and time period where relevant. Avoid dramatic wording that is not supported by the chart. Do not treat the dataset's labels as clinically validated unless source documentation establishes that validity.

Finding record template

Finding ID; question; chart/worksheet; metric; population and filters; observed pattern; numeric evidence; interpretation; alternative explanations; limitations; and reviewer. Add a screenshot or workbook reference so the finding can be checked.

Interpretation rules

A difference between groups may reflect sample composition, missingness, measurement choices, or confounding variables. An association does not prove causation. A dashboard may reveal a pattern but does not establish a mechanism, diagnosis, or appropriate treatment.

Limitations to assess

Dataset source and collection method; representativeness; sample size and subgroup sizes; field definitions; missing data; measurement error; selection bias; possible confounders; lack of temporal ordering; changing definitions; and whether the data is synthetic, de-identified, or drawn from a particular population. State only limitations that apply or are genuinely unknown, and explain how each affects interpretation.

Responsible communication

Use neutral, non-stigmatizing language. Avoid implying that a person's lifestyle or demographic group determines an outcome. Do not publish identifiers. Do not present exploratory risk labels as individual-level medical advice. Include a clear statement that the project is not a substitute for professional clinical assessment.

Conclusion template

Write a conclusion that answers the project's stated analytical questions, names the evidence used, acknowledges uncertainty, and identifies one or two realistic next steps. If a chart does not provide clear evidence, state that no firm conclusion can be drawn from the current data.

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References and Acknowledgements

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Dataset citation

Enter the dataset's exact title, author or organization, publication/update date, hosting location, URL or DOI, access date, license, and any citation instructions. Do not cite a guessed source. If the dataset was provided by an instructor or internal source, follow the permitted acknowledgement and confidentiality rules.

Technical references

Cite official documentation for Tableau features actually used, relevant data-cleaning tools, programming libraries, and website hosting. Include version numbers where known. Do not list a tool as used merely because it is mentioned in a template.

Research references

If the report explains medical or epidemiological concepts, use credible primary research or authoritative public-health sources and ensure the cited evidence supports the statement. Clearly separate background literature from observations made in the project's dataset.

Contribution statement

Describe who prepared the data, created Tableau worksheets, assembled the dashboard/story, built the website, tested the project, and wrote documentation. Ensure contributions and authorship are accurate and approved by the team.

Citation quality checklist

Every external factual claim has a citation; each citation resolves to the referenced source; URLs are accessible; the dataset license is respected; citations are consistent; and acknowledgements do not expose private information.

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Demonstration Script and Run of Show

Detailed project document · Prepared for project completion and review

Project reference information

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Suggested run of show

Opening and objective (about 30–45 seconds); data source and preparation (about 60 seconds); dashboard overview (2–3 minutes); story walkthrough (1–2 minutes); methodology and validation (about 60 seconds); limitations and conclusion (30–45 seconds). Adjust duration to the actual event requirements.

Opening script

“This project is Health360 Clinical & Healthcare Risk Analysis. It explores a healthcare-related dataset through Tableau visualizations. I will show the dataset scope, how the dashboard is organized, how to navigate the story, and what limitations should be considered when interpreting the results.” Replace this script with your natural speaking style and verified details.

Dashboard walkthrough

Begin with the dashboard title and scope. Explain the measures and groups shown by each major view. Demonstrate one filter and explain which charts respond. Point out a specific verified observation, including the measure and comparison. Avoid claiming a result until it has been checked against the workbook.

Story walkthrough

Explain why the story sequence was chosen. For each story point, state what the chart shows, what question it addresses, and what cannot be concluded. Move through the actual story rather than describing views that are not present.

Demo readiness checklist

Confirm internet access; open the website and direct Tableau links; check signed-out access; set browser zoom; close unrelated tabs; prepare a backup PDF or screenshots; ensure no sensitive data is visible; rehearse transitions; and have the dataset source and report available for questions.

Fallback plan

If an embedded view spins, open the direct Tableau view. If the public view is unavailable, show an approved screenshot/export and clearly disclose that the live view is not loading. Do not claim that the live interaction was demonstrated if it was not.

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Project Viva and Q&A — Detailed Preparation

Detailed project document · Prepared for project completion and review

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Project purpose

Q: What is Health360? A: An exploratory healthcare-related data-visualization project that organizes a dataset into Tableau worksheets, a dashboard, and a story. Add the exact analytical focus after checking the published workbook. Q: What problem does it solve? A: It makes patterns easier to explore than a raw table, while not claiming to replace clinical judgment.

Dataset questions

Q: What is the dataset size? A: Project notes describe 5,000 records and 17 columns; verify this against the actual file. Q: What are the columns? A: Answer from the source data dictionary, not memory or guesses. Q: How was the data obtained? A: Provide the exact source, retrieval date, license, and permitted-use terms. Q: How were missing values handled? A: Explain the actual field-level policy and its rationale.

Tableau questions

Q: Why Tableau? A: It supports interactive visual exploration, filters, dashboards, and stories. Q: Why these chart types? A: Explain the question and encoding for each actual worksheet. Q: How did you validate metrics? A: Describe independent calculations and reconciliation tests that were actually performed. Q: How do filters work? A: Demonstrate their real scope and reset behavior.

Analytical reasoning

Q: Does correlation prove causation? A: No. Observational group differences can be affected by confounding, selection, measurement, and missing data. Q: Is a risk category clinically validated? A: Only if the source/method has documented clinical validation; otherwise describe it as exploratory or project-defined. Q: Can it diagnose an individual? A: No, not as an educational visualization project.

Deployment questions

Q: How is the website hosted? A: The project website has been recorded as https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/; confirm it is live at review time. Q: Where is the source

code? A: https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git. Q: What if Tableau embeds do not load? A: Test each view directly, confirm public permissions, inspect network/browser restrictions, and verify the embed URL.

Ethics and limitations

Q: How do you protect privacy? A: Follow source permissions, remove/protect identifiers, minimize data, and verify that public release is authorized. Q: What are the limitations? A: State the actual limitations of source, coverage, definitions, missingness, and methodology. Q: What would you improve? A: Give realistic, evidence-based next steps rather than claiming features that have not been implemented.

Completion and review record

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Snowflake Data Platform Documentation

Detailed project document · Prepared for project completion and review

Project reference information

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Scope caution	This is an educational data-visualization project. It is not a clinically validated diagnostic, prognostic, or treatment system.

Applicability statement

Use this document only if Snowflake was actually used in the Health360 implementation. If the project uses only a local dataset and Tableau, mark Snowflake-specific implementation sections “Not used / Not applicable.” Do not imply that a Snowflake warehouse exists if it does not.

If Snowflake was used

Record account/environment name without secrets; database, schema, tables, stages, file formats, warehouses, roles, and integration points. Explain why the platform was selected and which components are part of the real workflow. Never include credentials, tokens, private keys, or sensitive connection strings.

Data ingestion

Document source file format, stage, load method, load date, file naming, error handling, and row-count reconciliation. Record the exact commands or pipeline definitions in a separate secure or version-controlled file, excluding secrets.

Data model

Describe table grain, primary/business keys if present, relationships, types, nullability, and retention. Avoid assigning key semantics to columns unless source documentation supports them. Record transformations and how the Snowflake data matches the Tableau data source.

Governance and security

Document least-privilege roles, data classification, masking, row-level controls if used, retention, and audit approach. Healthcare-related data may be sensitive; public access should be approved, not assumed.

Validation and cost

Record query results, load errors, row counts, performance observations, warehouse usage, and cost controls. If no Snowflake component was used, state that the project's data preparation and Tableau connection occurred through the actual alternative workflow.

Completion and review record

Prepared by: _____ Date: _____

Reviewed by: _____ Status: Draft / Reviewed / Approved

Document-control note: Update this file when the dataset, workbook, published views, or website changes. Record version/date and keep claims tied to evidence. This detailed document uses known project context and explicitly flags details that must be verified from the actual source files.

Snowflake SQL and Data Validation Log

Detailed project document · Prepared for project completion and review

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Applicability

This log is conditional. If Snowflake was not used, mark the document “Not applicable — Snowflake was not part of the implemented Health360 workflow.” Do not invent SQL execution evidence.

SQL inventory

For each executed query, record query ID, file or worksheet, purpose, source table, filters, expected output, actual output, run date, and reviewer. Keep secrets out of the document. Record the software/environment version when available.

Validation patterns

Potential checks include row counts, null counts, duplicate counts, allowed-category checks, range checks, and reconciliation between source and transformed tables. Use only queries actually run. Where example SQL is added, label it as an example and not as execution evidence.

Reconciliation

Compare record counts at each pipeline stage. Explain expected row loss or expansion caused by filtering or joins. For aggregates, compare a small sample with an independent calculation and document rounding rules.

Issue log

For each failure record severity, observed error, likely cause, corrective action, rerun result, and remaining risk. If an issue is unresolved, disclose it in the final report.

Evidence checklist

Attach query text or repository path, execution timestamp, result summary, and validation reviewer. If no Snowflake work was performed, keep the SQL inventory empty and explain the actual data pipeline elsewhere.

Completion and review record

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Tableau Workbook, Dashboard, Story, and Website Handoff

Detailed project document · Prepared for project completion and review

Project reference information

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Scope caution	This is an educational data-visualization project. It is not a clinically validated diagnostic, prognostic, or treatment system.

Asset inventory

The Health360 workflow has been described as containing nine Tableau worksheets/visualizations and a dashboard, with a story referred to as Patient Risk & Lifestyle Insights. Confirm the final workbook's actual sheet names and story title. Record workbook filename, author, version, data source, last modified date, and publication environment.

Worksheet register

Create one row for each of the nine worksheets: exact name; analytical question; dimensions; measures; aggregation; chart type; filters; calculation dependencies; validation reference; and screenshot/page. Do not substitute imagined worksheet titles for the actual workbook names.

Dashboard handoff

Record dashboard title, size/device layout, contained worksheets, filter scope, actions, navigation, color/format rules, and known limitations. Test every filter and action. Check whether titles, legends, axes, and tooltips remain correct after filtering.

Story handoff

Record exact story title, story-point order, each point's title and supporting worksheet, annotations, filter state, and narrative purpose. Confirm that each story point matches the published version and that the direct URL opens.

Published links and website

Project repository: https://github.com/en25cs3010918-ctrl/Health360-Healthcare-Risk.git. Project website: https://en25cs3010918-ctrl.github.io/Health360-Healthcare-Risk/. Enter the final direct Tableau dashboard URL and story URL separately after verifying each one. Do not assume the links are identical. Record access settings and whether the views work in a signed-out browser session.

Embed troubleshooting

If a view spins indefinitely: open the direct Tableau URL; confirm the workbook/view is published and accessible; check network/VPN/private DNS/content blockers; test another browser; inspect whether the embed path exactly matches the published view; ensure the correct Tableau loader or supported iframe pattern is used; and test each view independently. Record the actual result rather than claiming the issue is fixed without retesting.

Final handoff checklist

Deliver the workbook/source file where permitted, source dataset or provenance reference, data dictionary, cleaning log, dashboard/story URLs, website source, test evidence, known issues, and report. Remove credentials and restricted records. Verify that a reviewer can open the intended public assets.

Completion and review record

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